Chapter 2 — Introduction to data collection

Chapter 1 defined what a dataset is

Learning objectives

- Define data collection as a systematic
- Name the three process traits that distinguish rigorous campaigns from ad hoc gathering

What is data collection?

- Hypotheses, or operational decisions can be answered with evidence rather than anecdote
- How sampling choices shape what later analysts can claim

The five-step collection plan

- A practical collection plan usually proceeds through five linked decisions
- Teams first define objectives, research, monitoring, or model training
- They identify sources, such as primary experiments and surveys versus secondary archives
- Next they select methods, execute collection with quality checks

Process traits of effective collection

- Effective collection programs share three process traits
- They are systematic: procedures are specified so another team could reproduce the protocol
- They are objective-oriented
- They are dynamic

Example 2.1 — Collection plan for a retail satisfaction study

- Example 2.1 — hands-on module
- Example 2.1 converts a vague request into a collectable specification
- A retailer wants to know why online shoppers abandon carts
- The plan might specify an objective to identify checkout friction
- Explore the chapter example module
- View files: `modules/chapter2/example1/`

Why collection design matters

- Collection quality constrains everything that follows
- Analysis cannot recover information that was never measured
- When protocols are designed well

Takeaways

- Data collection is systematic and tied to objectives, not opportunistic gathering
- A five-step plan links purpose, sources, methods, execution, and storage
- Clarity about purpose and source comes before tool selection

Next

- Complete the quiz for this clip
- Continue to the next clip on sources of data, primary, secondary